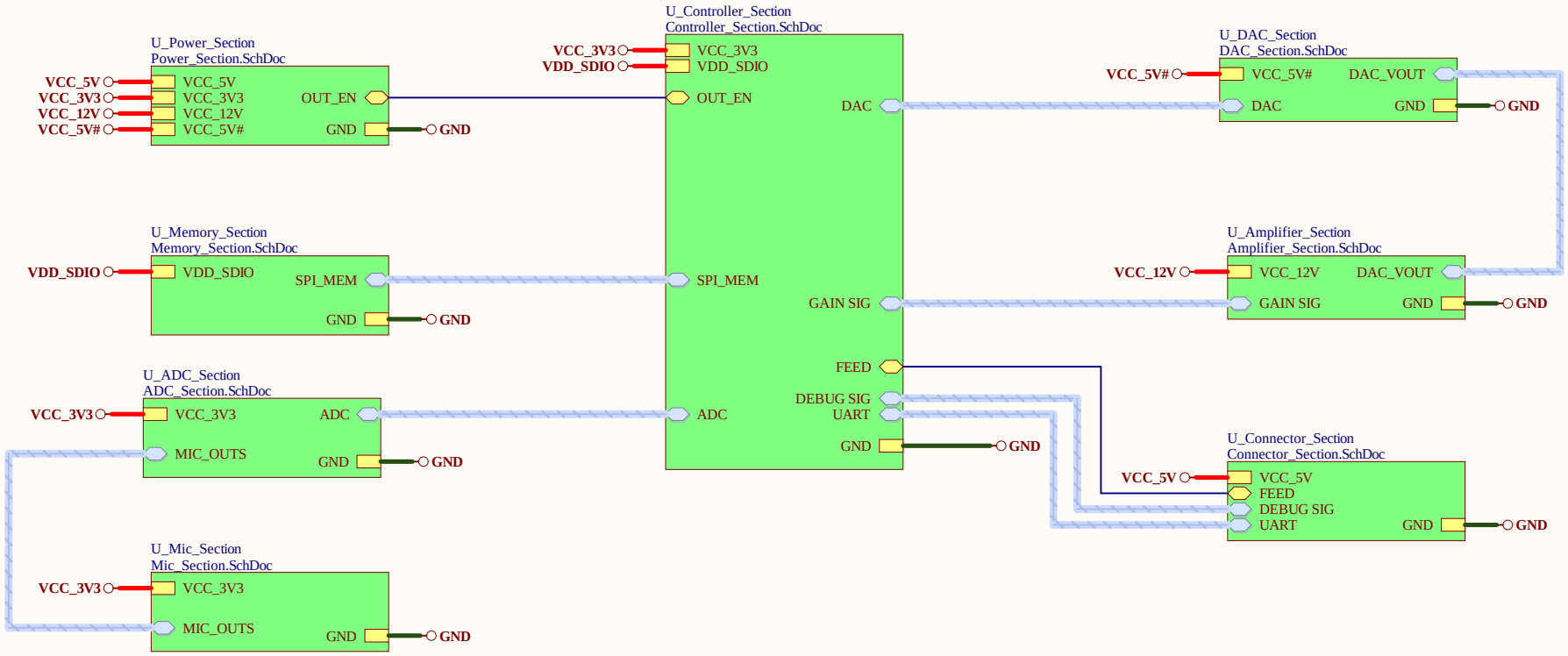
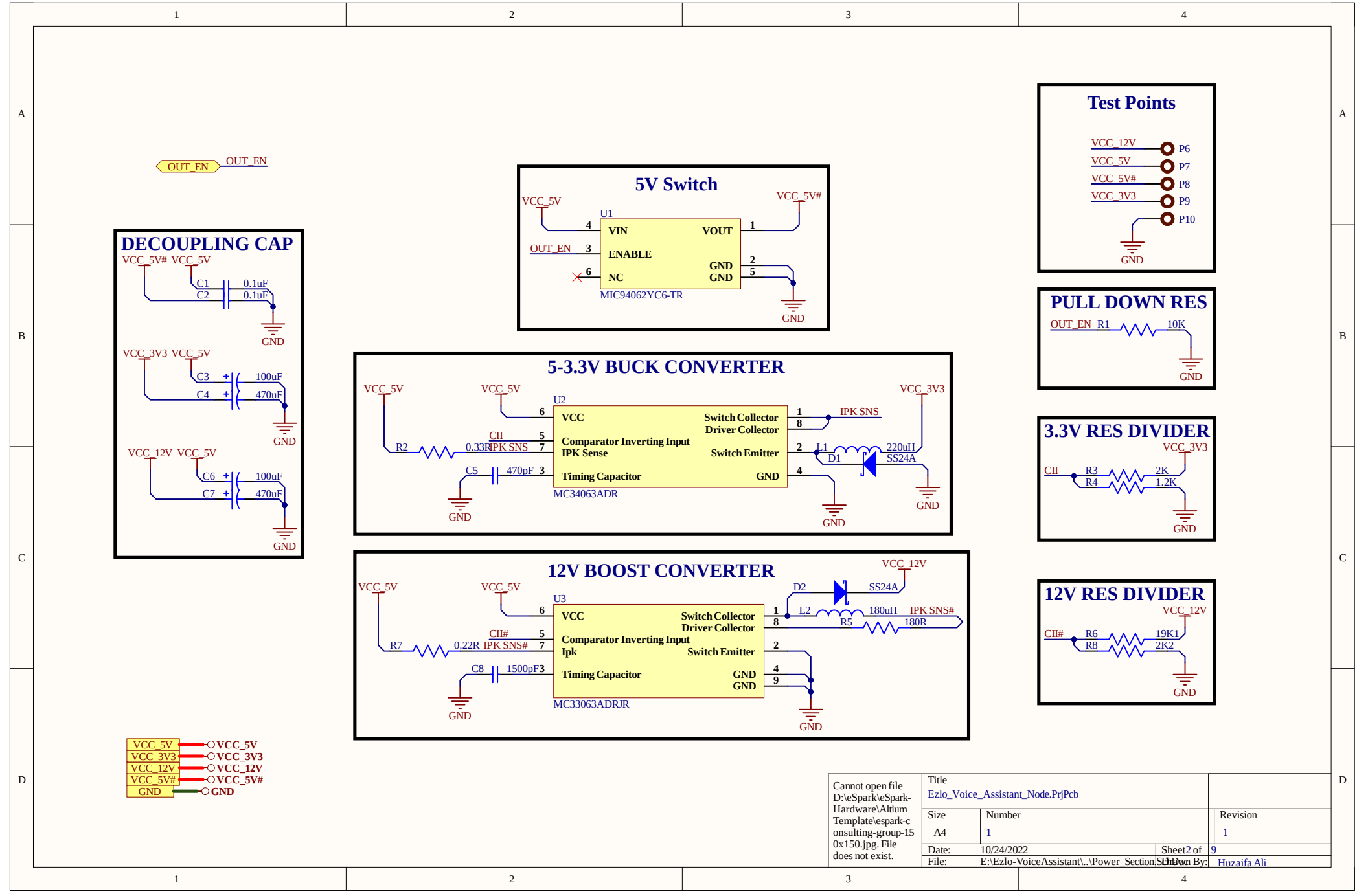
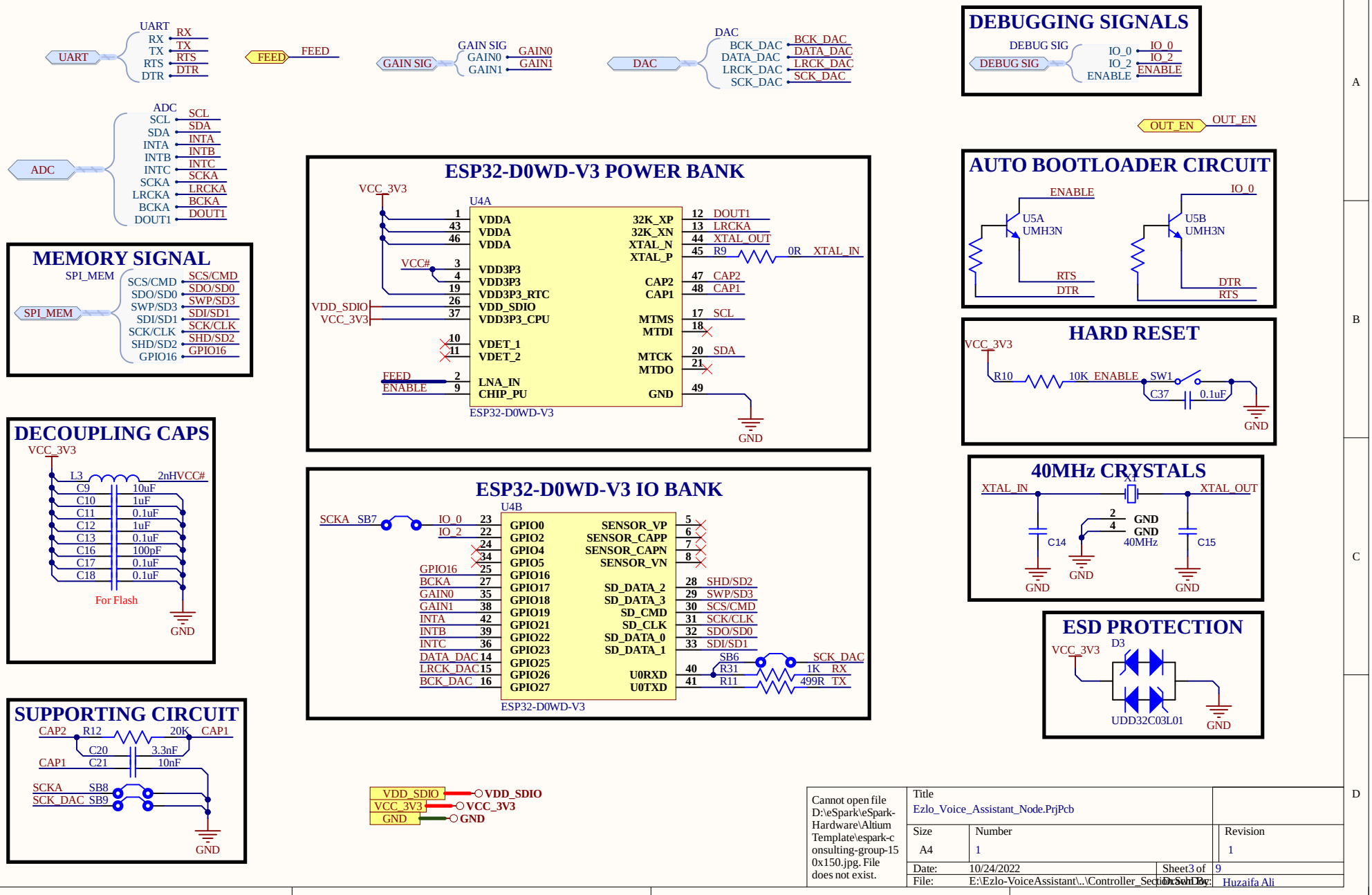


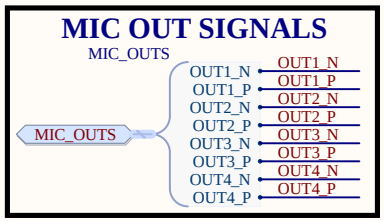
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	Size: A4	Number: 1	Revision: 1
	Date: 10/24/2022	Sheet 1 of 9	
	File: E:\Ezlo-VoiceAssistant\..\Top_Sheet.SchDoc	Drawn By: Huzaifa Ali	

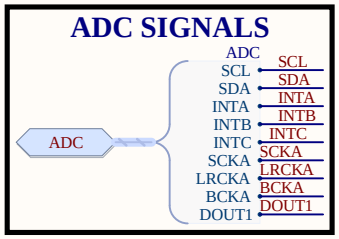


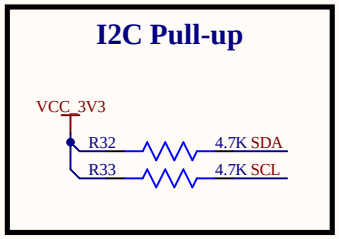
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	Size: A4	Number: 1	Revision: 1
	Date: 10/24/2022	Sheet 2 of 9	
	File: E:\Ezlo-VoiceAssistant\..\Power_Section.SchDoc	Drawn By: Huzaifa Ali	

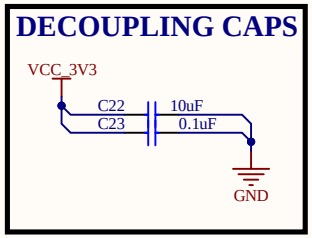


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	Size: A4	Number: 1
	Revision: 1	
	Date: 10/24/2022	Sheet 3 of 9
	File: E:\Ezlo-VoiceAssistant\..\Controller_Section.SchDoc	Drawn By: Huzaifa Ali

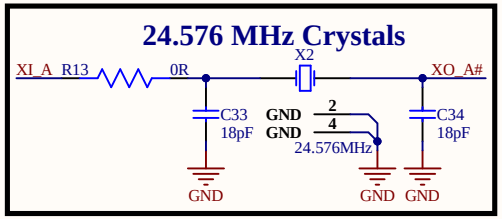


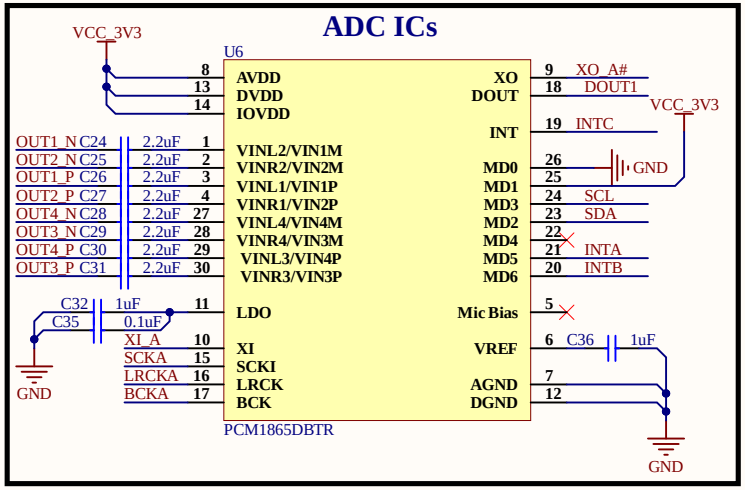






A minimum value of 2.2 µF is recommended in Figure 7 for codecs, which may have a very low input impedance at some PGA gain settings.
Datasheet page: 10

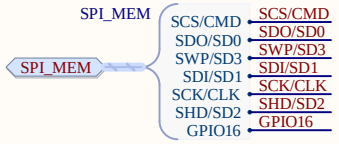




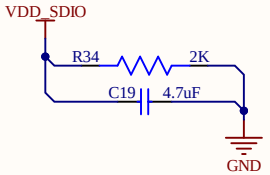


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	Size: A4	Number: 1	Revision: 1
	Date: 10/24/2022	Sheet 4 of 9	
	File: E:\Ezlo-VoiceAssistant\..\ADC_Section.SchDoc	Drawn By: Huzaifa Ali	

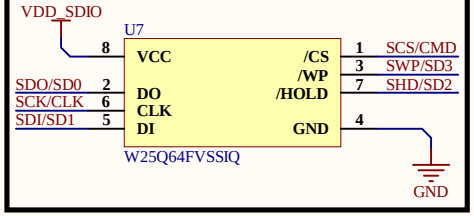
MEMORY SIGNAL



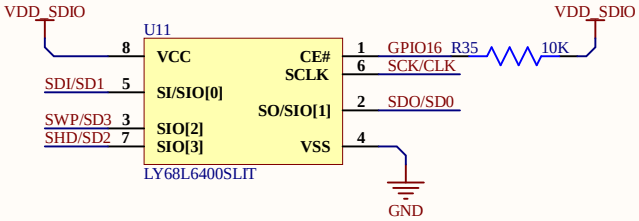
SUPPORTING CIRCUIT



64 Mbit SPI FLASH



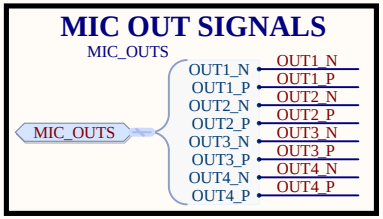
64 Mbit PSRAM



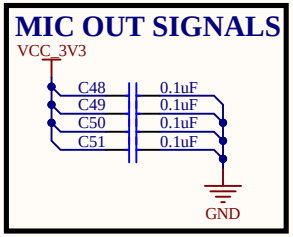


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_Assistant_Node.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 10/24/2022	Sheet 5 of 9	
	File: E:\Ezlo-VoiceAssistant\..\Memory_Section.SchDoc	Drawn By: Huzaifa Ali	

MIC OUT SIGNALS

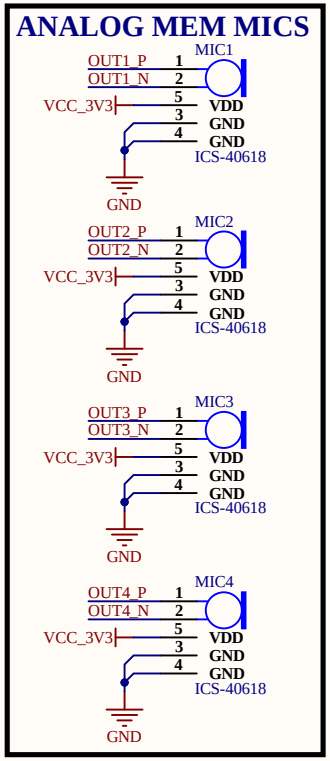


MIC OUT SIGNALS





ANALOG MEM MICS



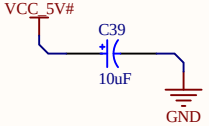
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	Size: A4	Number: 1	Revision: 1
	Date: 10/24/2022	Sheet 6 of 9	
	File: E:\Ezlo-VoiceAssistant\..\Mic_Section.SchDoc	Drawn By: Huzaifa Ali	



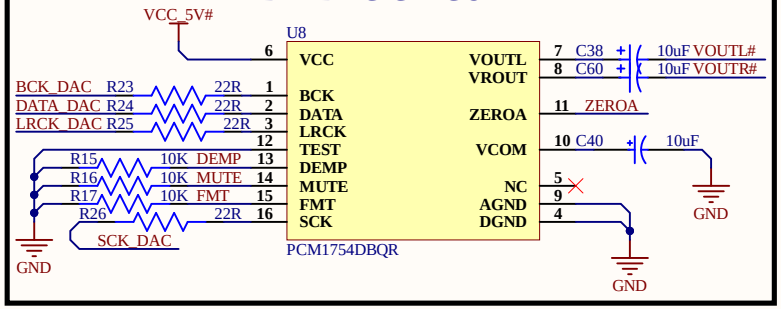


Input impedance of amplifier is 15K
when GAIN0-=0 and GAIN1=1
fC = 1/(2π × C × R)
Cutt of frequency of low pass filter is 10Hz
Range of sound signals= 20Hz-20KHz

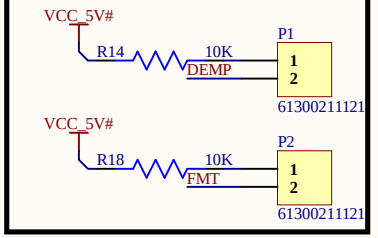
DECOUPLING CAP



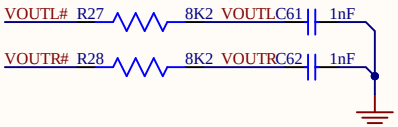
24 BIT DAC CIRCUITRY



FMT & DEMP CONTROL

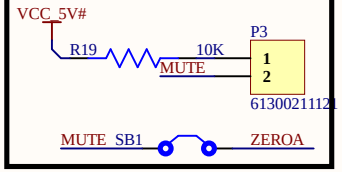


Supporting Circuitry



MUTE OFF - LOW
MUTE ON - HIGH
FMT LOW = 16– to 24–bit, I2S format
FMT HIGH =16–bit right–justified
DEMP HIGH = OFF
DEMP LOW = ON

MUTE CONTROL

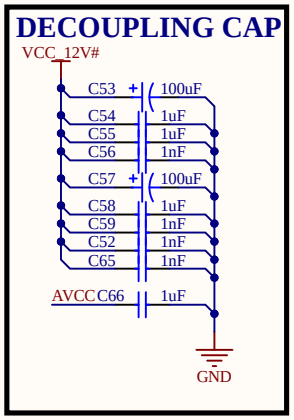


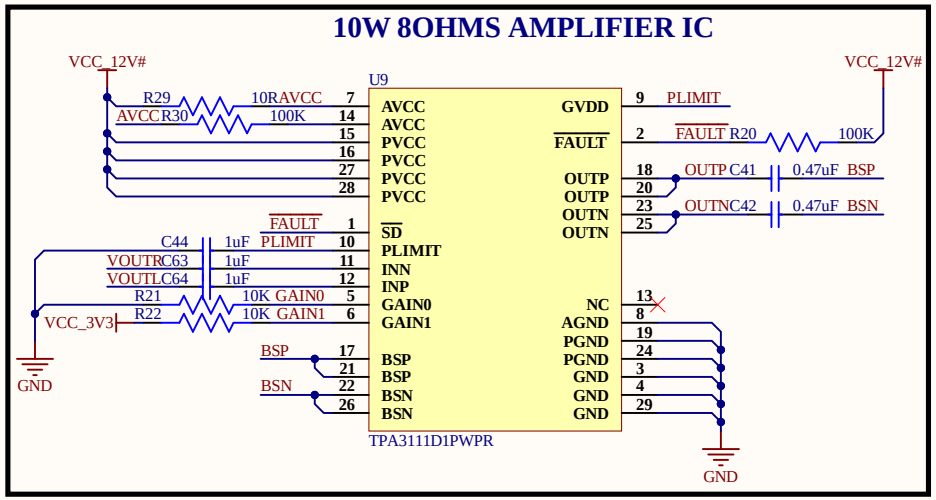


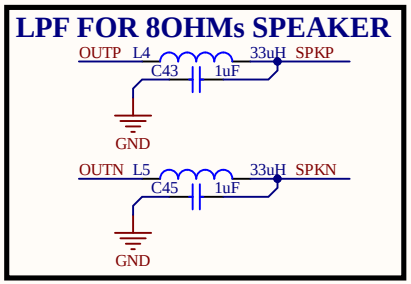
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	Size: A4 / Number: 1	Revision: 1
	Date: 10/24/2022 / Sheet 7 of 9	
	File: E:\Ezlo-VoiceAssistant\..\DAC_Section.SchDoc / Drawn By: Huzaifa Ali	

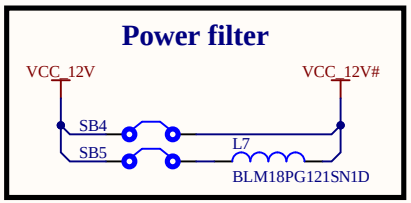


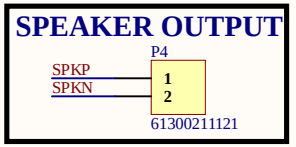


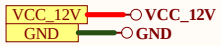








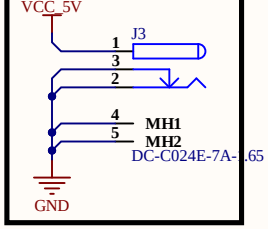




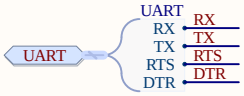
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	Size: A4	Number: 1	Revision: 1
	Date: 10/24/2022	Sheet 8 of 9	
	File: E:\Ezlo-VoiceAssistant\..\Amplifier_Section.SchDoc	Drawn By: Huzaifa Ali	



INPUT JACK



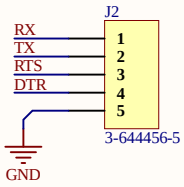
PROGRAM INPUTS



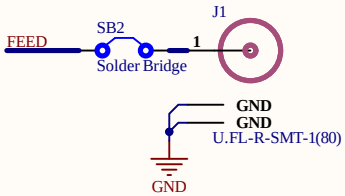
DEBUGGING SIGNALS



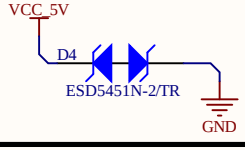
PROGRAMMING HEADER



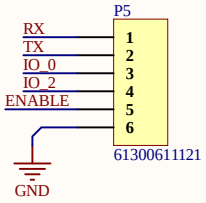
UFL CONECTOR EXTERNAL ANTENNA



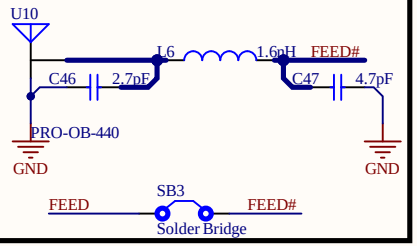
ESD PROTECTION



DEBUG CONNECTOR



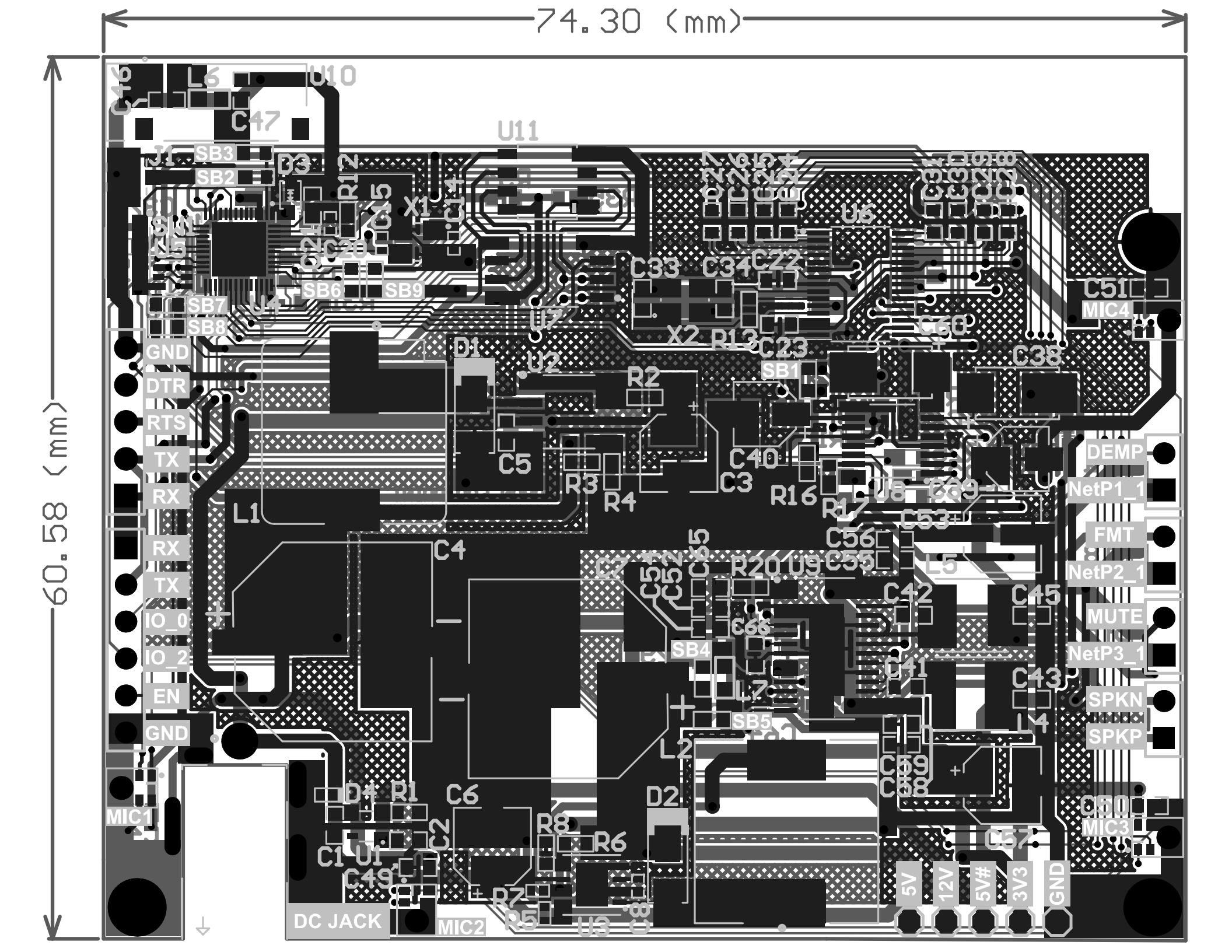
ON-BOARD PCB ANTENNA





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Title	Ezlo_Voice_Assistant_Node.PrjPcb		
Size: A4	Number: 1		Revision: 1
Date: 10/24/2022	Sheet 9 of 9		
File: E:\Ezlo-VoiceAssistant\..\Connector_Section.SchDoc	Drawn By: Huzaifa Ali		


74.30 (mm)
60.58 (mm)
U10
L6
C47
U11
SB3
SB2
D3
R12
C5
X1
C3
C2
C20
SB6
SB9
SB7
SB8
U4
U7
C33
C34
C22
X2
R13
C23
SB1
C60
C38
C51
MIC4
GND
DTR
RTS
TX
RX
RX
TX
IO_0
IO_2
EN
GND
D1
U2
R2
C5
R3
R4
C40
C3
R16
C39
C53
C56
C55
L5
C42
C45
C66
SB4
C41
C43
L7
SB5
L2
C4
L1
C6
D2
R8
R6
C1
U1
C49
C2
R7
R5
U3
C8
MIC1
MIC2
DC JACK
DEMP
NetP1_1
FMT
NetP2_1
MUTE
NetP3_1
SPKN
SPKP
C50
MIC3
C57
5V
12V
5V#
3V3
GND